Amazon E-Commerce Analytics

Project Overview:

This project analyzes a simulated Amazon E-Commerce dataset containing 10,000 transactions to identify business insights related to sales, customer behavior, product performance, returns, and delivery operations. Using Excel, SQL, and Power BI, the raw data was cleaned, transformed, and visualized into an interactive dashboard for business reporting and decision-making.

The project demonstrates end-to-end data analytics skills including data cleaning, SQL querying, KPI analysis, dashboard development, and insight generation aligned with Data Analyst and Business Intelligence roles.

Objectives:

- Analyze sales revenue trends across categories, brands, and time periods.
- Understand customer purchasing behavior, preferred payment methods, and geographic distribution.
- Evaluate product performance using ratings, review counts, and subcategory revenue.
- Examine delivery operations, including shipping times, delay patterns, and order status distribution.
- Measure return rates across categories and assess estimated profitability.
- Develop an interactive Power BI dashboard to visualize KPIs and business insights.
- Demonstrate end-to-end data analytics capabilities using Excel, SQL, and Power BI.

Dataset:

The dataset used in this project is a simulated Amazon E-Commerce transactional dataset containing 10,000 records and multiple business-related attributes. The data includes information related to customers, products, orders, payments, ratings, returns, and delivery operations.

Key dataset fields include:

- Order and customer details
- Product category, subcategory, and brand information
- Sales revenue, quantity, and pricing metrics

- Customer ratings and review counts
- Payment methods and order status
- Shipping dates, delivery times, and return indicators

The dataset was cleaned and transformed using Excel and SQL before being analyzed and visualized in Power BI to generate business insights and performance dashboards.

Tools & Technologies:

- **Excel** — Data cleaning, preprocessing, and initial exploration
- **SQL (MySQL)** — Data querying, transformation, and analytical calculations
- **Power BI** — Interactive dashboard development and data visualization
- **Power Query** — Data transformation and modeling within Power BI

Key Features:

- End-to-end data analytics workflow using Excel, SQL, and Power BI
- Data cleaning and preprocessing for accurate analysis
- SQL-based exploratory data analysis and KPI generation
- Interactive Power BI dashboard with multiple analytical views
- Sales, customer, product, delivery, and return analysis
- KPI tracking including revenue, orders, ratings, return rate, and shipping performance
- Business insights and recommendations derived from data trends
- Professional reporting and visualization for decision-making support

Key Insights:

Sales Insights

- Total revenue reached ₹99.41M from 10,000 orders.
- Electronics was the top-performing category with ₹66.17M revenue.
- February recorded the highest sales, while September had the lowest.
- Samsung emerged as the leading revenue-generating brand.

Customer Insights

- Payment methods were evenly distributed across customers.
- Cash on Delivery was the most preferred payment option.
- Order volumes remained consistent throughout the year.
- Major metro cities showed similar purchasing activity.

Product Insights

- Average product rating stood at 3.92/5.
- Headphones, Mobile, and Laptop were the highest-performing subcategories.
- Electronics dominated overall product revenue contribution.
- Beauty and Clothing categories generated the lowest revenue.

Delivery Insights

- Average shipping time was 3.20 days.
- A significant number of orders experienced delivery delays.
- Nearly 30% of orders remained in transit.
- Delivery analysis highlighted operational inefficiencies in fulfillment.

Returns & Profit Insights

- Overall return rate was 10.51%.
- Electronics contributed the highest return volume.
- Estimated profit was approximately ₹24.85M.
- Reducing return rates could significantly improve profitability.

Data Cleaning:

- Removed duplicate and inconsistent records from the dataset.
- Handled missing and null values across key columns.
- Standardized data formats for dates, prices, ratings, and categories.
- Corrected data type mismatches for numerical and transactional fields.
- Cleaned unwanted characters and validated text-based columns.
- Verified data accuracy before SQL analysis and Power BI visualization.

SQL Analysis – Queries & Business Logic:

A total of 16 structured SQL queries were developed across multiple business domains to analyze sales performance, customer behavior, product trends, delivery operations, and profitability. The queries were written using aggregation, grouping, filtering, sorting, and rounding techniques to generate meaningful business insights.

Sales Analysis:

- Calculated total revenue and estimated profit.

  Total Revenue:

```sql
SELECT ROUND(SUM(final_price), 2) AS total_revenue FROM amazon_ecomm;
-- Result: ₹ 99413864.31
```

Total Estimated Profit:

```
SELECT ROUND(SUM(estimated_profit), 2) AS total_estimated_profit FROM
amazon_ecomm;
-- Result: ₹ 24853478.79
```

- Analyzed revenue contribution by category and brand.

Revenue by Category:

```
SELECT category, ROUND(SUM(revenue), 2) AS revenue
FROM amazon_ecomm
GROUP BY category
ORDER BY revenue DESC;
-- Result:
'Electronics', '66170066.40'
'Home', '14679324.62'
'Sports', '11348685.32'
'Beauty', '3918395.27'
'Clothing', '3297392.70'
```

- Identified monthly sales trends and top-performing brands.

Monthly Sales Trend:

```
SELECT MONTH(purchase_date) AS month,
       ROUND(SUM(revenue), 2) AS revenue
FROM amazon_ecomm
GROUP BY month
ORDER BY month;
-- Result:
```

1	8298362.14
2	7518925.72
3	9073609.07
4	8147285.89
5	8439900.34
6	7813192.56
7	8352698.27
8	8712589.81
9	7331441.64
10	8068972.80
11	8364518.04
12	9292368.03

Top 10 Brands by Revenue:

```
SELECT brand, ROUND(SUM(revenue), 2) AS revenue
FROM amazon_ecomm
GROUP BY brand
ORDER BY revenue DESC
LIMIT 10;
-- Result:
'Samsung', '9149897.25'
'Boat', '8891013.43'
'Adidas', '8750256.36'
'Puma', '8713032.72'
'Sony', '8460265.88'
'HP', '8081193.47'
'Nike', '8032014.76'
'H&M', '7985250.73'
'Apple', '7972603.10'
'Lenovo', '7831193.77'
```

Customer Analysis:

- Evaluated customer purchasing patterns and payment preferences.

  Most Used Payment Method:

```
SELECT payment_method, COUNT(*) AS total_orders
FROM amazon_ecomm
GROUP BY payment_method
ORDER BY total_orders DESC;

-- Result:

Credit Card         2558
Cash on Delivery    2501
Debit Card          2479
UPI                 2461
```

- Identified top cities based on total order volume.

  Top Cities by Orders:

```
SELECT location, COUNT(*) AS order_count
FROM amazon_ecomm
GROUP BY location
ORDER BY order_count DESC;

-- Result:

Chennai     2023
Hyderabad   2021
Mumbai      1992
Delhi       1990
Bangalore   1973
```

Product Analysis:

- Analyzed product ratings and review performance.

  Highest Rated Products:

```
SELECT product_id, category, subcategory, rating
FROM amazon_ecomm
ORDER BY rating DESC
LIMIT 10;

--Result:

'P64254', 'Beauty', 'Makeup', '5'
'P59308', 'Electronics', 'Mobile', '5'
'P59941', 'Home', 'Decor', '5'
'P79604', 'Sports', 'Fitness', '5'
'P64381', 'Electronics', 'Laptop', '5'
'P52906', 'Beauty', 'Skincare', '5'
'P17592', 'Electronics', 'Mobile', '5'
'P78021', 'Home', 'Decor', '5'
'P52604', 'Electronics', 'Laptop', '5'
'P87278', 'Clothing', 'Kids', '5'
```

- Identified highly rated products and most reviewed brands.

  Most Reviewed Brands:

```
SELECT brand, SUM(review_count) AS total_ratings
FROM amazon_ecomm
GROUP BY brand
ORDER BY total_ratings DESC;

--Result:
Lenovo   28644
Adidas   28563
Samsung  28178
Sony     27717
HP       26881
Puma     26805
Zara     25904
H&M      25733
Nike     25639
Boat     25433
LG       24936
Apple    24511
```

Delivery Analysis:

- Measured average shipping time and delivery efficiency.

Average Shipping Time:

```
SELECT ROUND(AVG(shipping_time_days), 2) AS avg_shipping_days FROM
amazon_ecomm;
-- Result: 3.20 days
```

- Examined delivery status distribution and delayed orders.

Delivery Status Distribution:

```
SELECT delivery_status, COUNT(*) AS total_orders
FROM amazon_ecomm
GROUP BY delivery_status
ORDER BY total_orders DESC;
-- Result:
'In Transit\r', '3000'
'Delivered\r', '2979'
'Delayed\r', '2969'
'Returned\r', '1051'
```

Returns & Profit Analysis:

- Calculated overall return rate and category-wise returns.

Return Rate:

```
SELECT ROUND(SUM(is_returned) * 100.0 / COUNT(*), 2) AS return_rate
FROM amazon_ecomm;
-- Result: 10.51%
```

Returns by Category:

```
SELECT category, COUNT(*) AS total_returns
FROM amazon_ecomm
WHERE is_returned = 1
GROUP BY category
ORDER BY total_returns DESC;
--Result:
'Clothing', '226'
'Beauty', '222'
'Sports', '209'
'Electronics', '207'
'Home', '187'
```

- Analyzed seller performance and estimated profitability by category.

Top Sellers by Revenue:

```
SELECT seller_id, ROUND(SUM(revenue), 2) AS revenue
FROM amazon_ecomm
GROUP BY seller_id
ORDER BY revenue DESC
LIMIT 10;
--Result:
'S2290', '159646.34'
'S9262', '141460.84'
'S6835', '140410.89'
'S6149', '131642.64'
'S8277', '127662.42'
'S7366', '122888.23'
'S5118', '120642.18'
'S1579', '118091.36'
'S8192', '117596.12'
'S7730', '116558.52'
```

Estimated Profit by Category:

```
SELECT category, ROUND(SUM(estimated_profit), 2) AS profit
FROM amazon_ecomm
GROUP BY category
ORDER BY profit DESC;
--Result:
'Electronics', '16542519.00'
'Home', '3669833.48'
'Sports', '2837173.87'
'Beauty', '979601.64'
'Clothing', '824350.80'
```

The SQL analysis helped transform raw transactional data into actionable insights that were later visualized through interactive Power BI dashboards.

Power BI Dashboard – Visualizations:

An interactive multi-page Power BI dashboard was developed to visualize key business metrics and analytical insights from the Amazon E-Commerce dataset.

Executive Overview:

- KPI Cards: Total Revenue, Total Orders, Average Rating, Return Rate, and Average Shipping Days
- Revenue by Category
- Monthly Sales Trend
- Top Brands by Revenue
- Orders by City

Customer Insights:

- Payment Method Distribution
- Customer Purchase Trends
- Orders by Location
- Return Analysis by Customer Segments

Product & Seller Analysis:

- Top Products by Revenue
- Top Sellers Performance
- Product Rating Distribution
- Review Count Analysis

Delivery & Operations Analysis:

- Delivery Status Distribution
- Average Shipping Time Analysis
- Delayed Orders Tracking
- Delivery Performance Metrics

The dashboard was designed using Power BI visuals, DAX measures, slicers, filters, and interactive charts to support business reporting and decision-making.

Amazon E-Commerce Analytics

Delivery Status: All

purchase_date: All

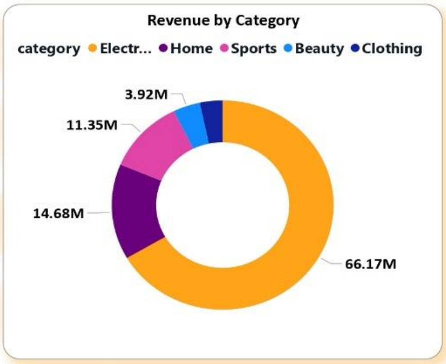
₹ 99.41M	10K	3.92	10.51	3.20
Total Revenue	Total Orders	Average Rating	Return Rate	Avg Shipping Days

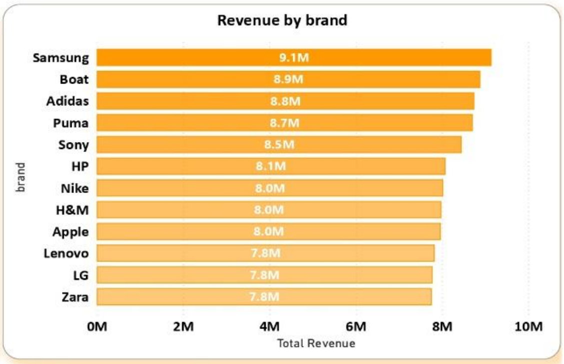
Brand: All

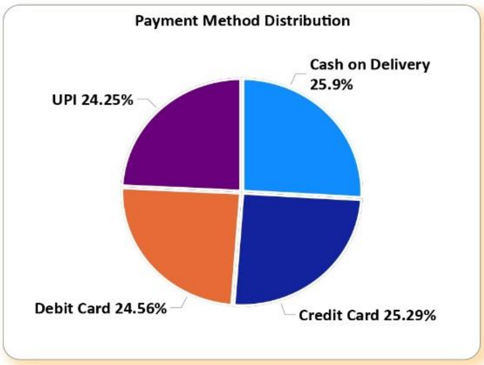
Category: All

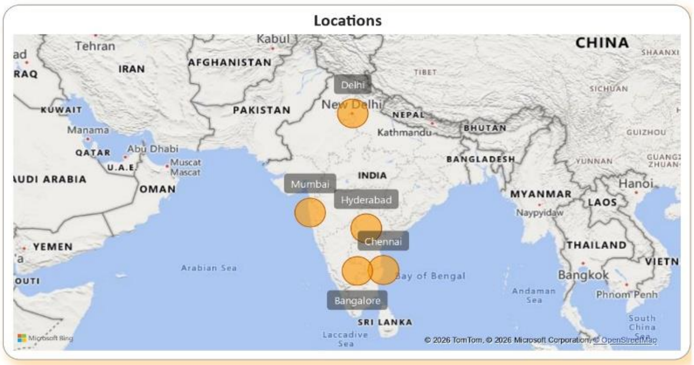
Location: All

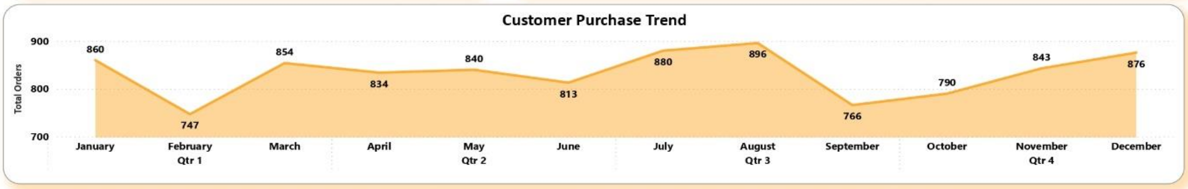
Payment Method: All

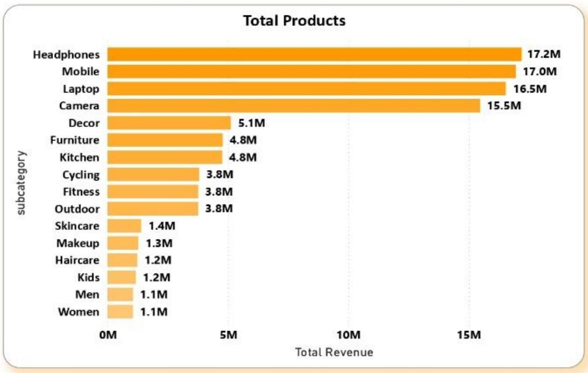


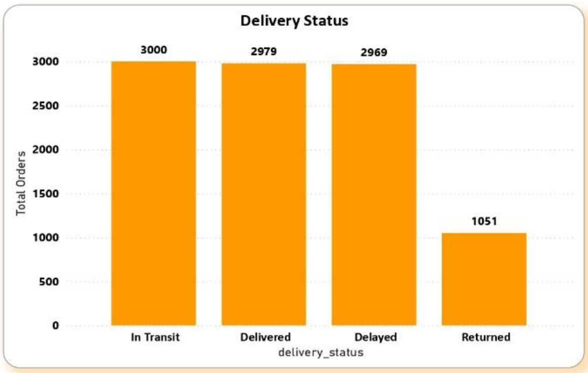












Recommendations:

- Focus on high-performing product categories and brands to maximize revenue growth.
- Improve delivery operations to reduce shipping delays and enhance customer satisfaction.
- Implement strategies to lower product return rates and improve profitability.
- Strengthen low-performing categories through targeted promotions and pricing strategies.
- Use customer purchasing insights to optimize marketing and payment experiences.
- Continuously monitor KPIs through dashboards for data-driven business decisions.

Conclusion:

This project successfully demonstrated an end-to-end data analytics workflow using Excel, SQL, and Power BI on a simulated Amazon E-Commerce dataset containing 10,000 records. Through data cleaning, SQL analysis, and dashboard visualization, meaningful insights were generated across sales, customer behavior, product performance, delivery operations, and profitability.

The project highlights practical skills in data analysis, business intelligence, KPI reporting, and dashboard development relevant to Data Analyst and BI Analyst roles.

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